

Simon Esbit, MD

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Education

2021 - 2026 MD, Medical School for International Health, Ben-Gurion University of the Negev

2021 - 2026 Certificate in Global Health, Ben-Gurion University of the Negev

2017 - 2019 BS, Neuroscience and Cognitive Science - Magna Cum Laude, University of Arizona

Experience

INDEPENDENT RESEARCHER | SEPTEMBER 2021-PRESENT

- Led research projects from study design through publication, including pilot studies, data collection, analysis, and manuscript writing.
- Managed research operations including grant writing and ethics submissions.
- Collaborated with clinicians and researchers across specialties and institutions to execute research projects: machine-learning-based diagnostics, global health, neurosurgery, psychiatry, genetics, and more.
- Translated complex clinical and research data into publications and actionable findings.

RESEARCH ASSISTANT II | HARVARD BRAIN TISSUE RESOURCE CENTER, MCLEAN HOSPITAL | SEPTEMBER 2019-AUGUST 2021

- Executed postmortem brain dissection protocols supporting neuropathological research.
- Automated worker productivity tracking (Python and VBA), reducing manual data entry and reporting errors.
- Performed complex SQL queries to integrate specimen and clinical databases.
- Managed lab procurement and financial operations for a \$12M NIH grant, including budgeting, equipment planning, vendor coordination, and invoice tracking.
- Interviewed prospective donors and their families. Coordinated brain donation with next-of-kin - obtaining medical records and regulatory documentation.
- Educated the public on neuroanatomy, mental health, and neurological and psychiatric disorders.

TEAM LEAD | SOCIAL COGNITIVE AND AFFECTIVE NEUROSCIENCE LAB, UNIVERSITY OF ARIZONA | JUNE 2017-SEPTEMBER 2019

- Led operations for a multi-phase, US-Department of Defense funded study modeling cognitive performance in combat medics under acute stress.
- Managed and trained research assistants to administer standardized neurocognitive, personality, emotional intelligence, and physiological assessments during acute stress and sleep deprivation protocols.
- Developed REDCap databases and Python scripts to process neurocognitive, physiologic, and behavioral data.
- Conducted statistical analyses (R), reported findings to the US-Department of Defense, and published research.
- Spearheaded study design and collaborated with IRB on protocol development, amendments, and human subjects compliance for experimental stress paradigms.

RESEARCH ASSISTANT I | WIENS LAB, UNIVERSITY OF ARIZONA | SEPTEMBER 2016-JUNE 2017

- Performed RAD-seq workflows (DNA extraction, digestion, PCR, ligation) for population genomic analysis of *Oedipodea* grasshopper species.
- Conducted spatial and climatic niche analyses (R): range map digitization and species distribution modeling to assess climatic constraints influencing divergence in sister species.

Awards

Pillars of Excellence Award McLean Hospital; Mass General Brigham 2020

Meritorious Poster Award American Speech-Language-Hearing Association 2018

Publications

- Ofri, M., Azriel, A., Kahati, L., Paran, N., **Esbit, S.**, Sufaro, Y., Barda, N. (2026). Predicting chronic subdural hematoma risk in elderly patients with mild traumatic brain injury. *Acta Neurochirurgica*. 168, 40. <https://doi.org/10.1007/s00701-026-06788-5>
- Williams, J. M., **Esbit, S. L.**, Ho, M. L., & Sidlow, R. (2025). Neurodevelopmental and Behavioral Phenotypes in 14q11.2 Microduplication Syndrome: A Case Report and Literature Review. *Cureus*, 17(8), e90735. <https://doi.org/10.7759/cureus.90735>
- Galindo, A. M., Azriel, A., Pasternak, D., **Esbit, S.**, Mudrik, A., Avraham, E., Sufaro, Y., & Meirovitz, A. (2025). Awake craniotomy vs general anesthesia for brain metastases in eloquent areas: neurological and survival outcomes. *Journal of clinical neuroscience*. 139, 111457. <https://doi.org/10.1016/j.jocn.2025.111457>
- **Esbit, S.**, Kaim, A., Kimhi, S., Bankauskaite, D., Baran, M., Baran, T., Cosciug, A., Eshel, Y., Dumbadze, S., Gabashvili, M., Jiglaui, G., Kaniasty, K., Koubova, A., Maricano, H., Matkeviciene, R., Maticescu, M., Nazarov, M., Teperik, D., Kochiashvili, N., & Adini, B. (2025). Hope and Distress: A Cross-Country Study Amid the Russian-Ukrainian War. *Stress and health : journal of the International Society for the Investigation of Stress*, 41(2), e70033. <https://doi.org/10.1002/smi.70033>
- Abu-Freha, N., Afawi, Z., Yousef, M., Alamor, W., Sanalla, N., **Esbit, S.**, & Yousef, M. (2025). A machine learning approach to differentiate stage IV from stage I colorectal cancer. *Computers in biology and medicine*, 191, 110179. <https://doi.org/10.1016/j.compbiomed.2025.110179>
- Abu-Freha, N., Alamour, O., Weissmann, S., **Esbit, S.**, Cohen, B., Gordon, M., Abu-Freha, O., El-Saied, S., & Afawi, Z. (2024). Ethnic Disparities in Chronic Diseases: Comparison Between Arabs and Jews in Israel. *The Israel Medical Association journal : IMAJ*, 26(11), 694–700. <https://www.ima.org.il/MedicineIMAJ/viewarticle.aspx?year=2024&month=12&page=694>
- **Esbit, S.**, & Sidlow, R. (2024). A Case of Beta-Propeller Protein-Associated Neurodegeneration With a Unique Truncating Variant in the WDR45 Gene and Uncommon Clinical and Radiologic Findings. *Cureus*, 16(4), e58127. <https://doi.org/10.7759/cureus.58127>
- Vanuk, J. R., Pace-Schott, E. F., Bullock, A., **Esbit, S.**, Dailey, N. S., & Killgore, W. D. S. (2022). Morning blue light treatment improves sleep complaints, symptom severity, and retention of fear extinction memory in post-traumatic stress disorder. *Frontiers in behavioral neuroscience*, 16, 886816. <https://doi.org/10.3389/fnbeh.2022.886816>
- Solomon, A., Alteber, Z., Bassan, D., Sharbi-Yunger, A., **Esbit, S.**, Tzehoval, E., & Eisenbach, L. (2022). On the development of a neoantigen vaccine for the prevention of Lynch Syndrome. *International journal of cancer*, 151(1), 107–119. <https://doi.org/10.1002/ijc.33971>

Manuscripts (in preparation)

- **Esbit, S.**, LaFollette, K., Satterfield, B., A., & Killgore, W. D. Cognitive reflection and disaster triage: performance in an online training environment. In preparation.
- **Esbit, S.**, LaFollette, K., Satterfield, B., A., & Killgore, W. D. Unmasking implicit bias: racing diffusion modeling reveals the effects of sleep deprivation on racial biases. In preparation.

Book Chapters (in preparation)

- Ho M. L., Sidlow R., Williams J., **Esbit S.** Pediatric head and neck vascular tumors and malformations. In: Eldaya R., ed. Vascular Malformations of the Head and Neck. *Neuroimaging Clinics*. In preparation.

Poster Presentations

- LaFollette, K., Satterfield, B. C., Lazar, M., **Esbit, S.**, Anlap, I., & Killgore, W. D. S. (2020, November). Chunking of sequence memories during high-stress sleep deprivation. 61st Annual Meeting of the Psychonomic Society, Virtual.

- LaFollette, K., Satterfield, B. C., Lazar, M., **Esbit, S.**, Anlap, I., & Killgore, W. D. S. (2020, June 13–17). Emotion regulation during sleep deprivation and repeated physiological stress: Implications for motor skill learning and production. 34th Annual SLEEP Meeting, Philadelphia, PA.
- Satterfield, B. C., Anlap, I., **Esbit, S.**, & Killgore, W. D. S. (2020, June 13–17). Corticotropin-releasing hormone receptor 1 gene polymorphism modulates cognitive flexibility following acute stress and total sleep deprivation. 34th Annual SLEEP Meeting, Philadelphia, PA.
- Burns, A., Shepard, C., Bullock, A., **Esbit, S.**, Alkozei, A., Satterfield, B., & Killgore, W. D. S. (2020, March 19–22). The association between life history strategy and anxiety is mediated by trait gratitude. Anxiety and Depression Conference, San Antonio, TX.
- **Esbit, S.**, Raygoza, D., Meinhausen, C., Dailey, N., & Killgore, W. D. S. (2020, February 5–8). Discrepancies between working memory and clustering strategies in total recall performance for mild traumatic brain injuries. Neuropsychology in the Era of Precision Medicine: Annual Conference of the International Neuropsychological Society, Denver, CO.
- **Esbit, S.**, Satterfield, B., LaFollette, K., Lazar, M., & Killgore, W. D. S. (2020, February 5–8). Biological sex differences and overriding misleading impulses. Neuropsychology in the Era of Precision Medicine: Annual Conference of the International Neuropsychological Society, Denver, CO.
- Meinhausen, C., **Esbit, S.**, Dailey, N., & Killgore, W. D. S. (2020, February 5–8). Self-initiated verbal recall strategies following mild traumatic brain injury. Neuropsychology in the Era of Precision Medicine: Annual Conference of the International Neuropsychological Society, Denver, CO.
- Anlap, I., **Esbit, S.**, Alkozei, A., Satterfield, B., & Killgore, W. D. S. (2020, February 5–8). The effects of gratitude on wellbeing are mediated by social support. Neuropsychology in the Era of Precision Medicine: Annual Conference of the International Neuropsychological Society, Denver, CO.
- **Esbit, S.**, Briemann, S., & Killgore, W. D. S. (2019, August 19–22). Exploration of emotional intelligence and self-perceived invincibility. Military Health System Research Symposium, Kissimmee, FL.
- Strong, M., **Esbit, S.**, LaFollette, K., Dailey, N., & Killgore, W. D. S. (2019, August 19–22). Big Five personality traits and how they relate to self-perceived invincibility. Military Health System Research Symposium, Kissimmee, FL.
- LaFollette, K., Satterfield, B. C., **Esbit, S.**, Lazar, M., & Killgore, W. D. S. (2019, August 19–22). Self-perceived invincibility is associated with greater cognitive flexibility. Military Health System Research Symposium, Kissimmee, FL.
- LaFollette, K., Satterfield, B. C., **Esbit, S.**, Lazar, M., & Killgore, W. D. S. (2019, August 19–22). Inadequate sleep quality and duration predicts disinhibited shooting on a “shoot/no-shoot” task. Military Health System Research Symposium, Kissimmee, FL.
- LaFollette, K., Satterfield, B. C., **Esbit, S.**, Lazar, M., Grandner, M. A., & Killgore, W. D. S. (2019, June 8–12). Negative mood and poor sleep are associated with altered moral reasoning under stress. 33rd Annual SLEEP Meeting, San Antonio, TX.
- LaFollette, K., Satterfield, B. C., **Esbit, S.**, Lazar, M., Grandner, M. A., & Killgore, W. D. S. (2019, June 8–12). The effects of prior at-home sleep duration on reversal-learning during a “shoot/no-shoot” task. 33rd Annual SLEEP Meeting, San Antonio, TX.
- **Esbit, S.**, Dailey, N., & Killgore, W. D. (2019, February 20–23). Making a list and checking it twice: Episodic verbal recall in mild traumatic brain injury. 47th Annual Conference of the International Neuropsychological Society: Embracing the Biopsychosocial Melting Pot, New York City, NY.
- **Esbit, S.**, LaFollette, K., Botello, R., Satterfield, B., Alkozei, A., & Killgore, W. D. (2019, February 20–23). High self-perceived adroitness: An altered perception of reality during sleep deprivation. 47th Annual Conference of the International Neuropsychological Society: Embracing the Biopsychosocial Melting Pot, New York City, NY.
- Forbeck, B., Dailey, N., **Esbit, S.**, & Killgore, W. D. (2018, November 15–17). Reduced information processing speed: A dynamic deficit in mild traumatic brain injury. 2018 American Speech-Language-Hearing Association: Revolutionary Learning, Evolutionary Practice, Boston, MA. * **Meritorious Poster Submission**